Formulas

Physicists calculate unknown quantities using formulas, in which known quantities are combined in specific ways. Formulas can be rearranged according to which quantity needs to be calculated. Here are some of the main formulas.

PHYSICS FORMULAS

Quantity	Description	Formula
Current	$\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
Voltage	current x resistance	$V = IR$
Resistance	$\frac{\text{voltage}}{\text{current}}$	$R = \frac{V}{I}$
Power	$\frac{\text{work}}{\text{time}}$	$P = \frac{W}{t}$
Time	$\frac{\text{distance}}{\text{velocity}}$	$t = \frac{d}{v}$
Distance	velocity x time	$d = vt$
Velocity	$\frac{\text{displacement (distance in a given direction)}}{\text{time}}$	$v = \frac{d}{t}$
Acceleration	$\frac{\text{final velocity} - \text{initial velocity}}{\text{time}}$	$a = \frac{v2\text{-}v1}{t}$
Force	mass x acceleration	$F = ma$
Momentum	mass x velocity	$p = mv$
Pressure	$\frac{\text{force}}{\text{area}}$	$P = \frac{F}{A}$
Density	$\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
Volume	$\frac{\text{mass}}{\text{density}}$	$V = \frac{m}{\rho}$
Mass	volume x density	$m = V\rho$
Area	length x width	$A = lw$
Kinetic energy	½ mass x square of velocity	$E_k = \frac{1}{2} mv^2$
Weight	mass x acceleration due to gravity	$W = mg$
Work done	force x distance moved in direction of force	$W = Fs$

The planets

This table gives some basic information on the planets of the Solar System plus the number of observed moons that orbit them. The inner planets have rocky surfaces, while the larger outer planets are mainly made of gases and ice.

PLANETS AND MOONS

Planet	Description	Number of known moons
Mercury	rock, metal	0
Venus	rock, metal	0
Earth	rock, metal	1
Mars	rock, metal	2
Jupiter	gas, ice, rock	63
Saturn	gas, ice, rock	62
Uranus	gas, ice, rock	27
Neptune	gas, ice, rock	14

Earth's vital statistics

Our planet is the largest rocky planet in the Solar System. Many of the units scientists use to measure the Universe are based on the size and motion of the planet.

Average diameter	12,756 km (7,928 miles)
Average distance from Sun: km (miles)	149.6 million (93 million)
Average orbital speed around Sun: km (miles)	29.8 km/s (18.5 mps)
Sunrise to sunrise (at the Equator)	24 hours
Mass	5.98×10^{24} kg
Volume	1.08321×10^{12} km^3
Average density (water = 1)	5.52 g/cm^3
Surface gravity	9.81 m/s^2
Average surface temperature	15°C (59°F)
Ratio of water to land	70:30